

CLEVELAND HARBOR, OH, USA

WMO: 997692

Lat: 41.542N

Lon: 81.637W

Elev: 689

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.4	9.3	N/A	N/A	N/A	N/A	N/A	N/A	32.3	27.0	29.1	27.3	8.9	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	86.9	N/A	83.9	N/A	81.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.1	26.2	23.3	DB	0.1	92.1	8.8	1.9	-6.3	93.5	-11.4	94.6	-16.4	95.7	-22.8	97.1	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	27.6	27.4	36.2	48.8	60.2	68.4	72.7	71.4	65.8	55.0	43.1	35.8	(6)	
	DBStd	18.17	11.19	10.92	10.06	9.33	8.61	6.45	5.35	4.75	6.72	7.52	8.31	8.95	(7)	
	HDD50	2647	700	637	448	134	10	0	0	0	0	32	239	447	(8)	
	HDD65	5803	1160	1054	894	495	203	36	4	5	69	322	657	904	(9)	
	CDD50	3065	5	3	18	98	325	551	703	662	473	187	32	8	(10)	
	CDD65	749	0	0	0	9	54	138	241	202	92	13	0	0	(11)	
	CDH74	4118	0	0	0	86	298	751	1494	949	493	47	0	0	(12)	
	CDH80	864	0	0	0	13	60	183	307	174	125	2	0	0	(13)	
(14) Wind	WSAvg	8.8	10.4	9.9	10.1	10.2	7.5	7.0	6.7	6.7	7.5	9.5	10.1	10.0	(14)	
(15) Precipitation	PrecAvg	40.3	3.0	2.3	2.8	3.5	3.7	3.8	4.2	3.5	3.9	3.5	2.9	3.1	(15)	
	PrecMax	52.5	6.1	5.5	6.0	5.8	7.3	8.0	9.3	7.8	9.0	6.8	5.0	7.1	(16)	
	PrecMin	28.2	1.5	0.9	0.6	1.7	1.4	0.8	1.3	1.3	1.2	1.5	1.2	1.6	(17)	
	PrecStd	5.7	1.1	1.0	1.1	1.1	1.7	1.7	1.9	1.4	1.7	1.2	1.0	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.5	65.5	70.5	82.2	85.9	89.5	89.6	90.0	89.2	78.7	66.3	63.7	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	56.2	54.4	64.6	76.9	81.2	84.9	86.0	84.4	83.9	74.5	62.9	58.8	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	49.5	47.7	59.3	71.1	77.3	81.6	83.3	80.8	79.4	71.1	59.8	54.4	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	42.9	41.1	52.7	65.9	73.3	78.3	81.1	78.7	76.2	67.3	56.4	49.0	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.8	12.8	14.6	14.1	12.0	11.0	10.7	12.2	12.6	11.4	10.2	(35)	
		MCDBR	18.3	20.0	22.9	24.8	20.2	17.3	14.4	14.3	16.7	17.1	17.1	16.6	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.306	0.335	0.359	0.397	0.433	0.451	0.451	0.436	0.388	0.369	0.328	0.309	(40)	
	taud		2.402	2.278	2.293	2.275	2.208	2.201	2.224	2.257	2.365	2.397	2.488	2.478	(41)	
	Ebn at Noon		267	274	281	277	268	263	262	262	268	259	258	257	(42)	
	Edh at Noon		25	34	37	41	45	45	44	41	35	30	23	22	(43)	
(44) All-Sky Solar Radiation	RadAvg		432	706	1034	1415	1702	1891	1896	1676	1333	847	531	347	(44)	
	RadStd		39	89	104	125	147	141	122	79	121	75	85	32	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	N/A

Nomenclature: See separate page